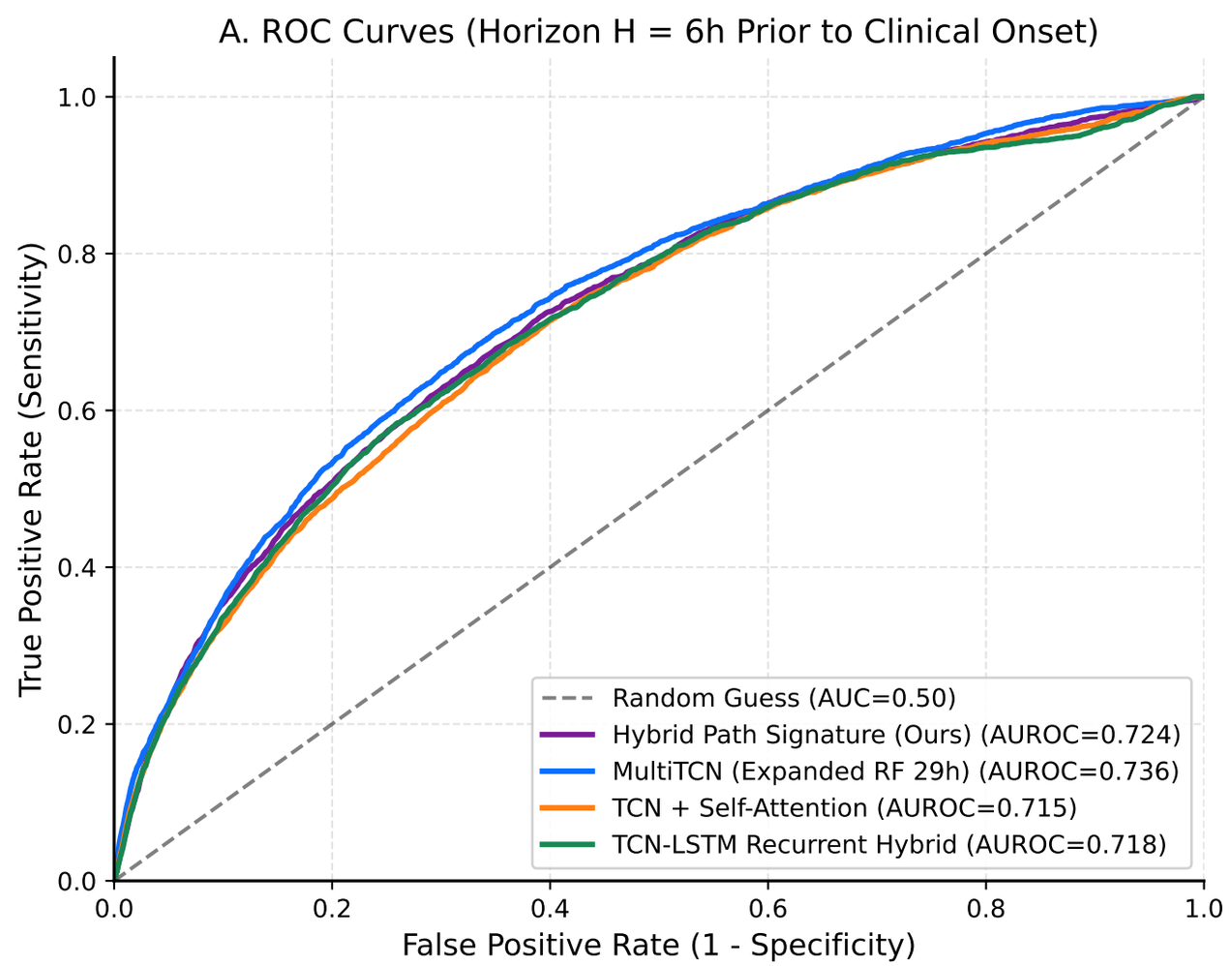
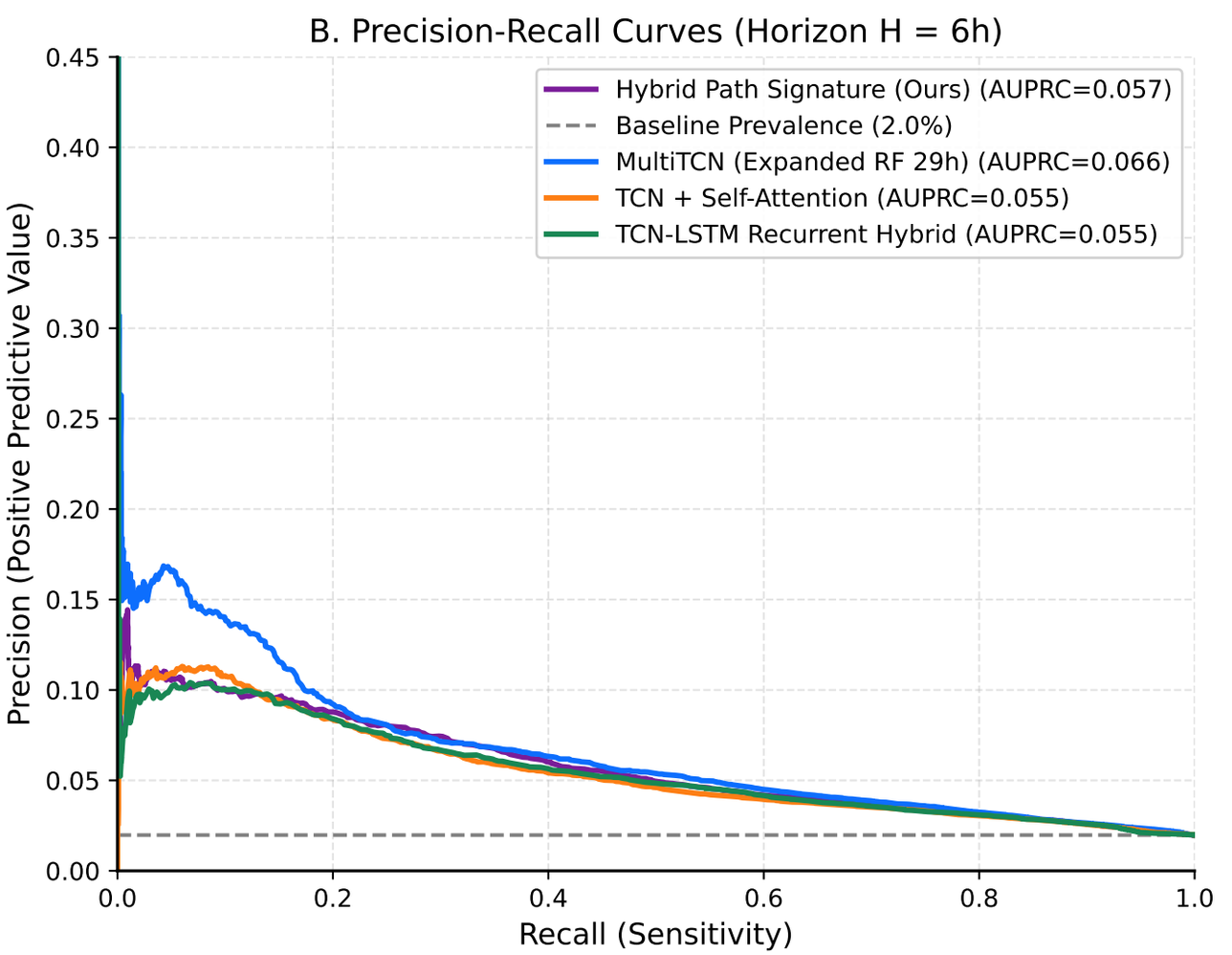
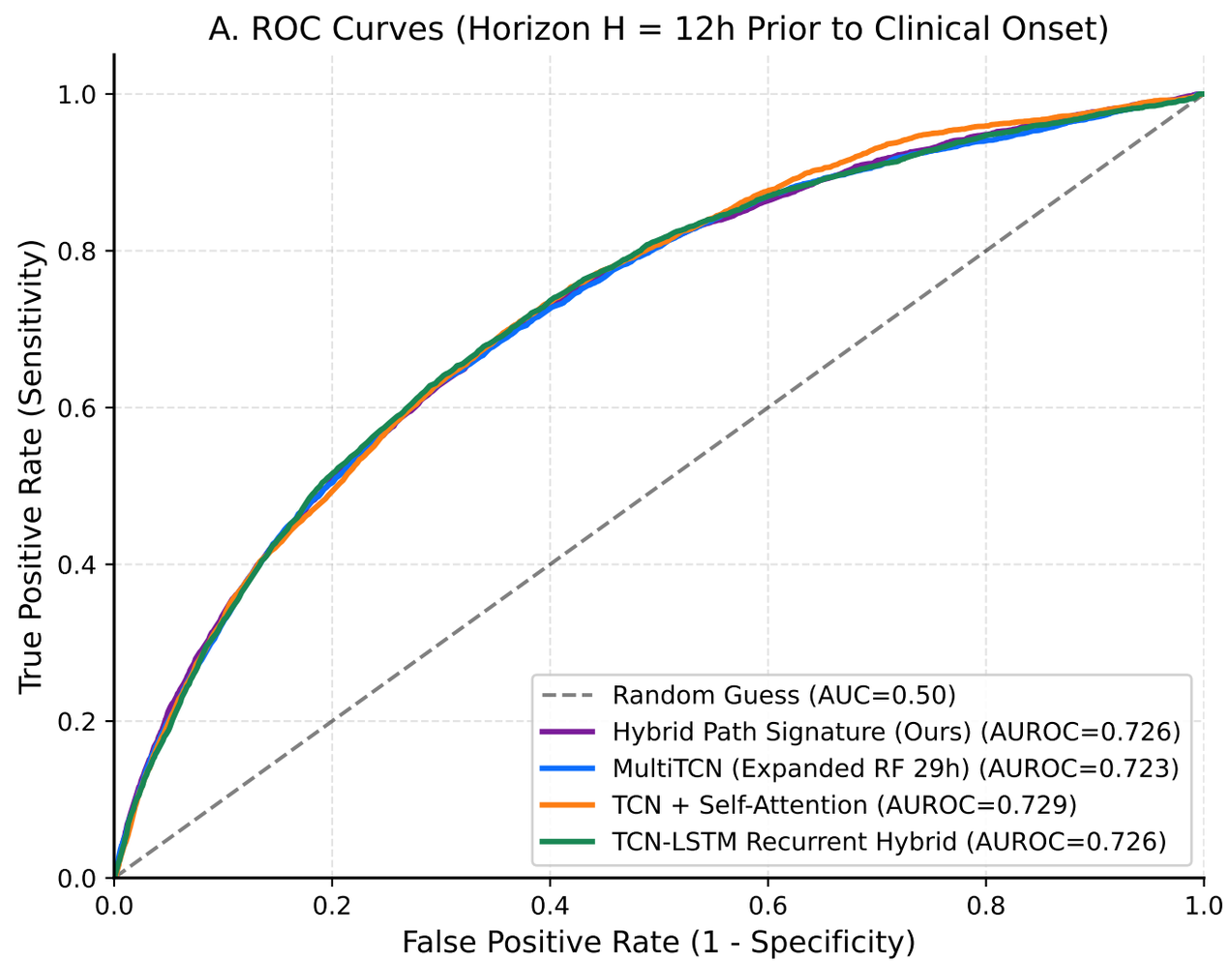
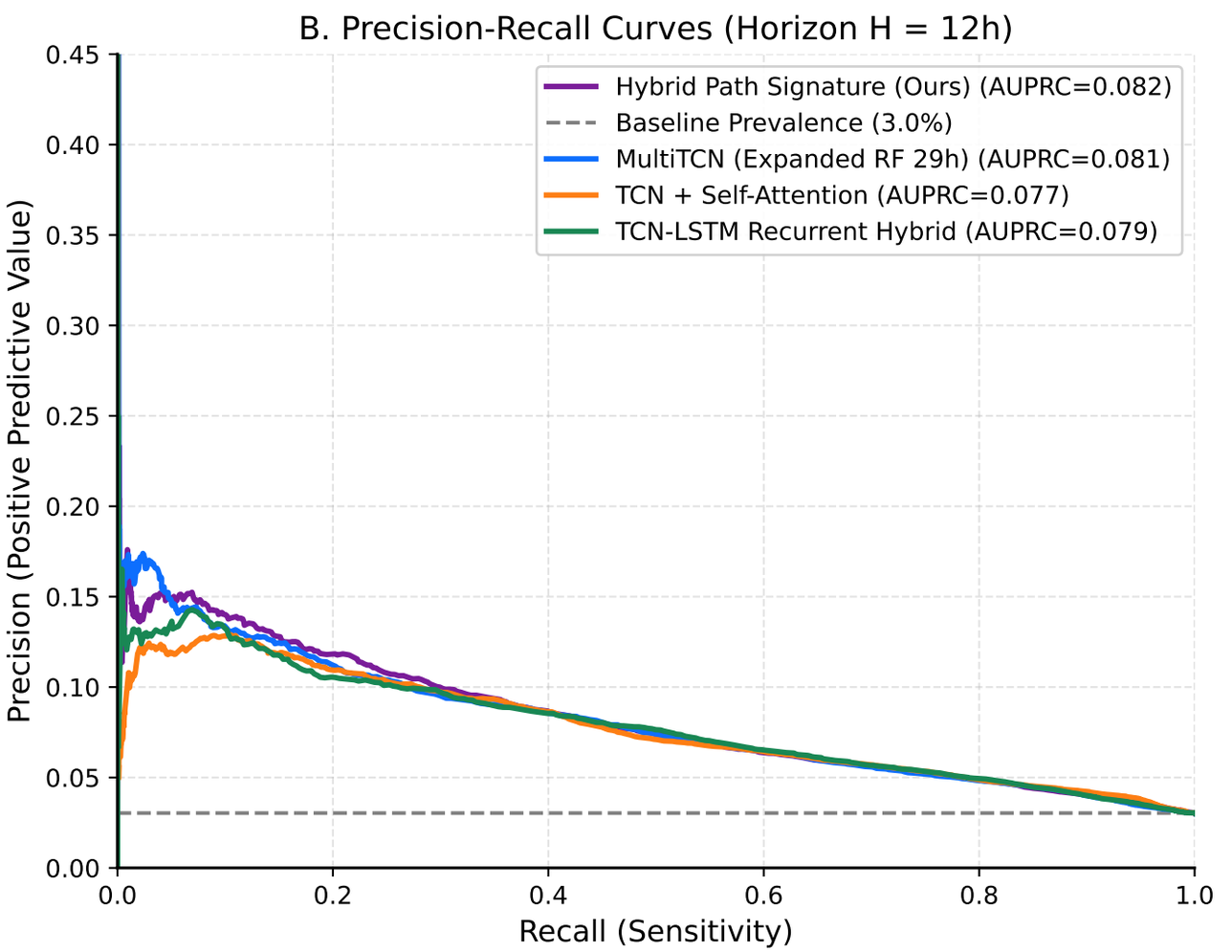

A. ROC Curves (Horizon H = 6h Prior to Clinical Onset)
True Positive Rate (Sensitivity)
False Positive Rate (1 - Specificity)
0.0
0.2
0.4
0.6
0.8
1.0
Random Guess (AUC=0.50)
Hybrid Path Signature (Ours) (AUROC=0.724)
MultiTCN (Expanded RF 29h) (AUROC=0.736)
TCN + Self-Attention (AUROC=0.715)
TCN-LSTM Recurrent Hybrid (AUROC=0.718)


B. Precision-Recall Curves (Horizon H = 6h)
Precision (Positive Predictive Value)
Recall (Sensitivity)
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
0.45
0.0
0.2
0.4
0.6
0.8
1.0
Hybrid Path Signature (Ours) (AUPRC=0.057)
Baseline Prevalence (2.0%)
MultiTCN (Expanded RF 29h) (AUPRC=0.066)
TCN + Self-Attention (AUPRC=0.055)
TCN-LSTM Recurrent Hybrid (AUPRC=0.055)


A. ROC Curves (Horizon H = 12h Prior to Clinical Onset)
True Positive Rate (Sensitivity)
False Positive Rate (1 - Specificity)
0.0
0.2
0.4
0.6
0.8
1.0
Random Guess (AUC=0.50)
Hybrid Path Signature (Ours) (AUROC=0.726)
MultiTCN (Expanded RF 29h) (AUROC=0.723)
TCN + Self-Attention (AUROC=0.729)
TCN-LSTM Recurrent Hybrid (AUROC=0.726)


B. Precision-Recall Curves (Horizon H = 12h)
Precision (Positive Predictive Value)
Recall (Sensitivity)
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
0.45
0.0
0.2
0.4
0.6
0.8
1.0
Hybrid Path Signature (Ours) (AUPRC=0.082)
Baseline Prevalence (3.0%)
MultiTCN (Expanded RF 29h) (AUPRC=0.081)
TCN + Self-Attention (AUPRC=0.077)
TCN-LSTM Recurrent Hybrid (AUPRC=0.079)